PROJECT REPORT · FEATURES

AEGIS AI SMART SURVEILLANCE

Turns ordinary CCTV cameras into a smart security team. It spots wanted people, guns, intruders and vehicles on its own, and alerts the control room within seconds.

HOW IT WORKS

Every video frame passes through 6 AI stages

Camera — CCTV · RTSP · USB

Detect — People · guns · cars

Track — ID for each one

Recognize — Faces · plates

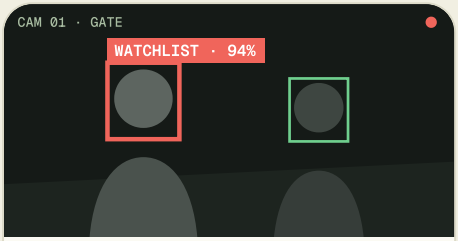
Behavior — Zones · falls

Analytics — Counts · heatmaps

Alert — Seconds later

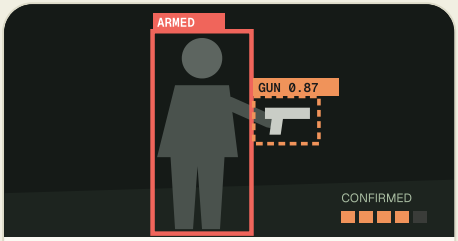
WHAT AEGIS CAN SEE

Each picture shows what the operator sees on screen



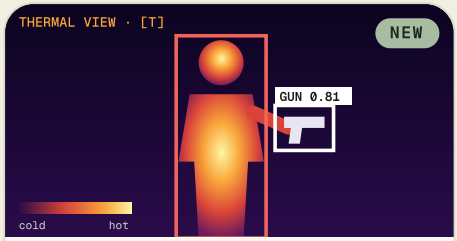
Criminal Face Match

Checks every face against the watchlist. A match draws a red box, sounds an alarm and alerts every channel.



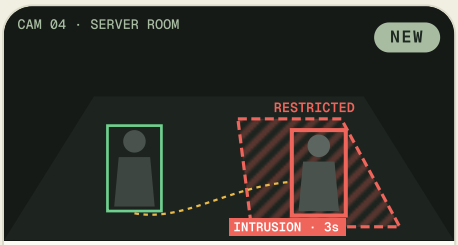
Gun & Knife Detection

Finds the weapon and links it to the person holding it. It must appear in several frames before an alert, so a single-frame glitch can't trigger one.



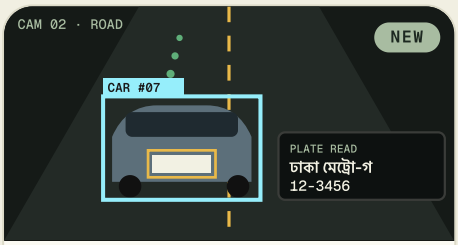
Thermal View

Press T to switch to a heat-map colour view. Person and gun boxes stay on top, so a weapon stands out in dark or busy scenes.



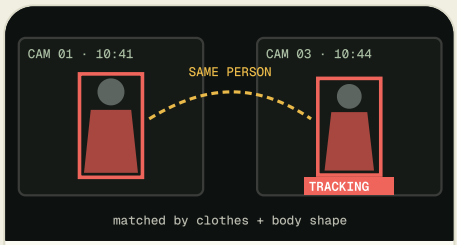
Restricted Area Entry

Admins draw no-go zones on the camera view. Anyone who enters, loiters, crowds in or crosses a line triggers an alert.



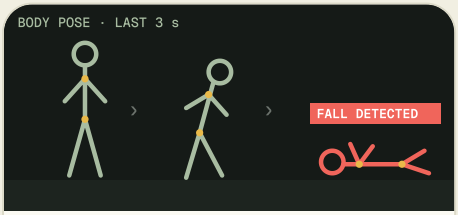
Car & Vehicle Tracking

Follows cars, bikes, buses and trucks. Reads number plates, including Bangla, and flags any vehicle on the watchlist.



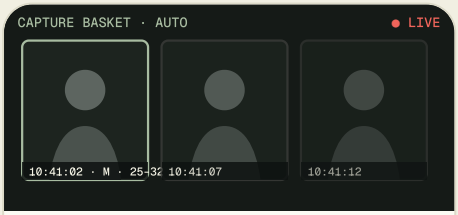
Cross-Camera Tracking

Remembers a suspect's clothes and body shape, then follows them from one camera to the next.



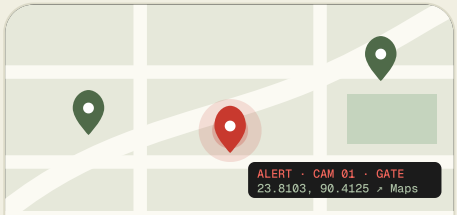
Fall & Stagger Detection

Reads body posture over 3 seconds and raises an alert when someone falls or staggers.



Capture Basket

While a suspect is on camera, Aegis saves a zoomed photo every few seconds, with an estimated gender and age.

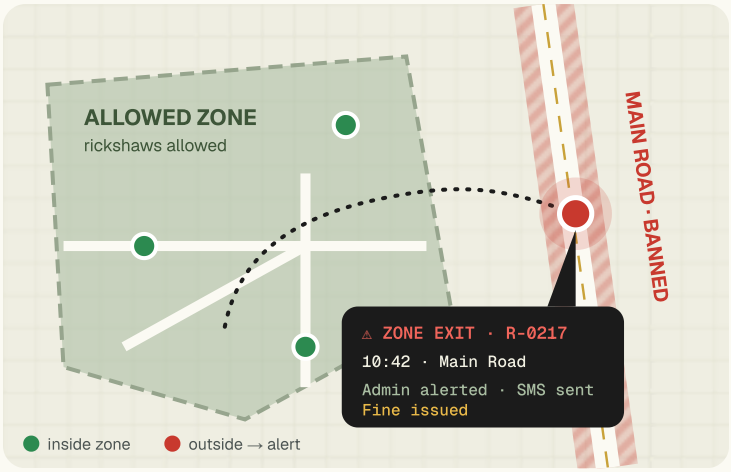


GPS Map

Every camera is pinned on a live map. An alert shows the exact spot and a Google Maps link for responders.

PROPOSED · SMART RICKSHAW TRACKING

GPS geo-fence with automatic fine

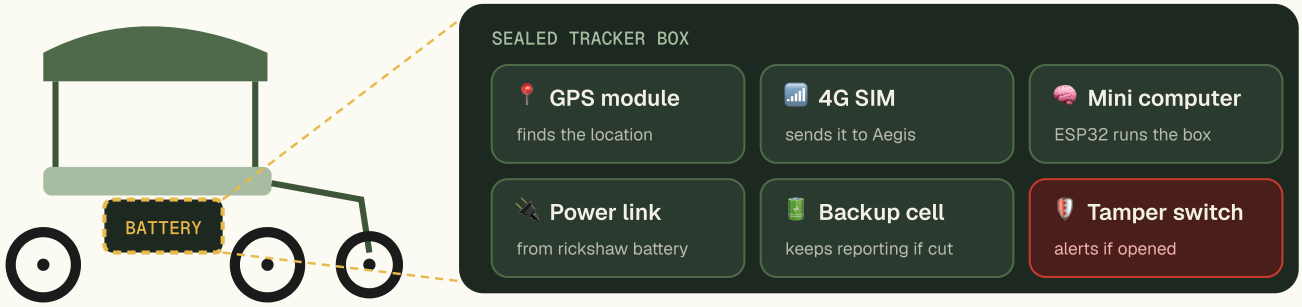


HOW IT WORKS

1. **Tracker in the battery**
   A small sealed box with GPS and a 4G SIM, powered by the rickshaw's own battery.
2. **Location every 10 seconds**
   It reports the rickshaw's position to the Aegis server over mobile data.
3. **Geo-fence check**
   Admins draw allowed zones and banned roads on the map. Aegis checks every new position against them.
4. **Exit = alert + fine**
   A red alert appears on the dashboard and live map, the owner gets an SMS, and a fine is issued automatically with the GPS trail as proof.
5. **Anti-tamper**
   An alert also fires if the box is opened, unplugged, or stops reporting.

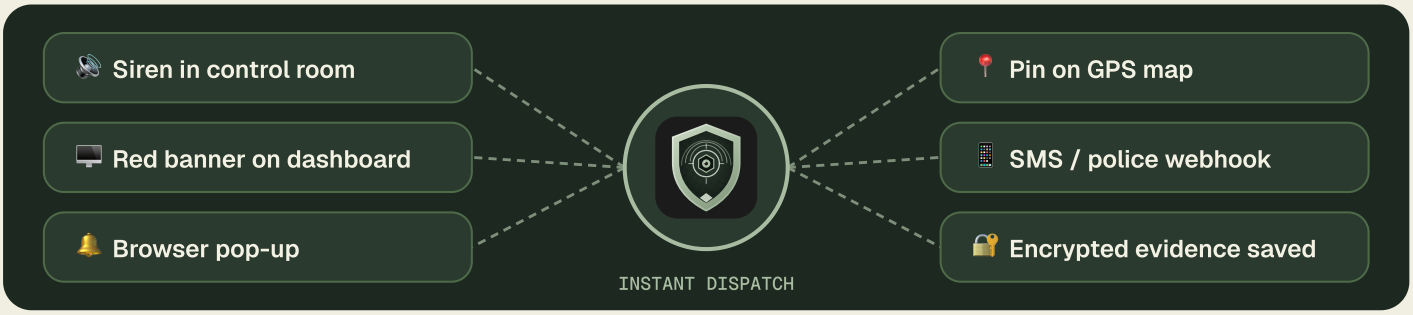
INSIDE THE RICKSHAW TRACKER

Hidden in the battery compartment, runs on the rickshaw's own power



ONE EVENT, EVERY CHANNEL

What happens the moment something is detected



ALERT LEVELS

Every event is ranked so the most serious ones come first

CRITICAL	HIGH	MEDIUM	INFO
Wanted face, gun, tracked suspect, watchlisted car	Restricted-zone entry, suspicious behaviour, fall	Loitering, crowding, line crossing	People counts, camera on/off, system status

THE CONTROL ROOM

Dashboard tools for operators and admins

Live Camera Wall	Zone Editor	Analytics	Incident Reports	Command Assistant	Secure Access
All feeds, full-screen, start/stop	Draw restricted areas	Counts, heatmaps, trends	One-click report	Ask in plain words	Roles, lockout, audit log